



NSW Education Standards Authority

2018 HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 1

**General
Instructions**

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions 11–14, show relevant mathematical reasoning and/or calculations

**Total marks:
70****Section I – 10 marks** (pages 2–5)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 6–13)

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 The polynomial $2x^3 + 6x^2 - 7x - 10$ has zeros α , β and γ .

What is the value of $\alpha\beta\gamma(\alpha + \beta + \gamma)$?

- A. -60
- B. -15
- C. 15
- D. 60

- 2 The acute angle between the lines $y = 3x$ and $y = 5x$ is θ .

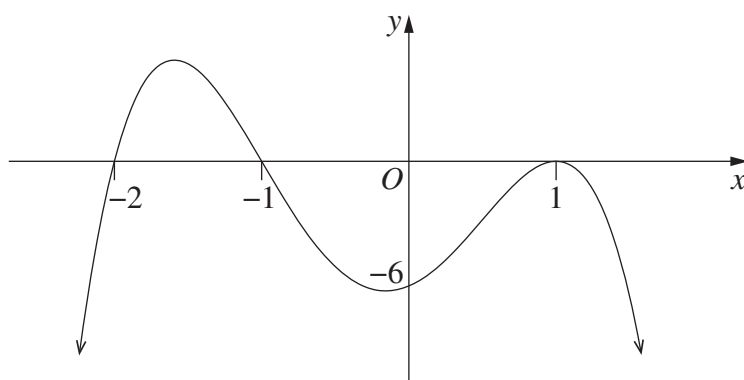
What is the value of $\tan \theta$?

- A. $\frac{1}{8}$
- B. $\frac{1}{7}$
- C. $\frac{1}{2}$
- D. $\frac{4}{7}$

- 3 What is the value of $\lim_{x \rightarrow 0} \frac{\sin 3x \cos 3x}{12x}$?

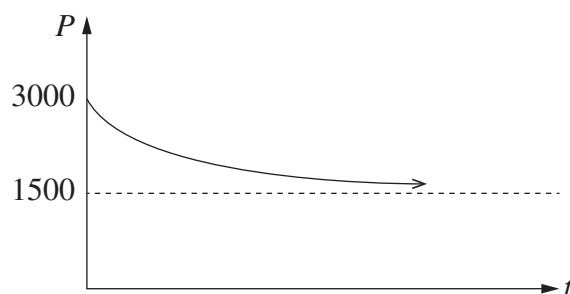
- A. $\frac{1}{4}$
- B. $\frac{1}{2}$
- C. $\frac{3}{4}$
- D. 1

- 4 The diagram shows the graph of $y = a(x + b)(x + c)(x + d)^2$.



What are possible values of a , b , c and d ?

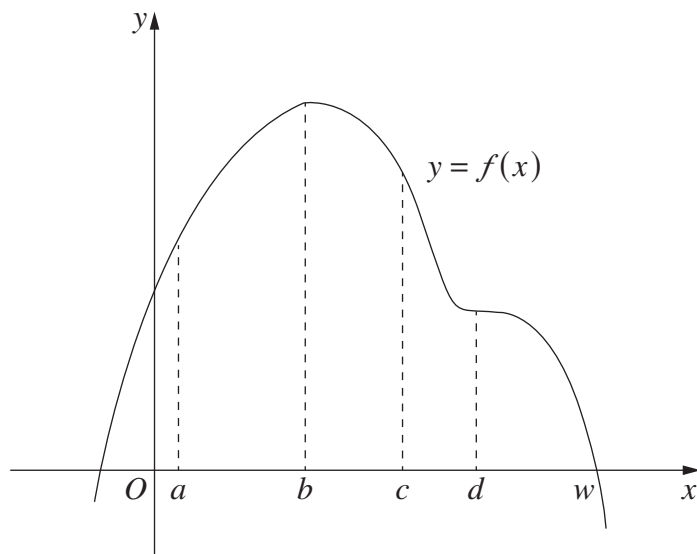
- A. $a = -6$, $b = -2$, $c = -1$, $d = 1$
 - B. $a = -6$, $b = 2$, $c = 1$, $d = -1$
 - C. $a = -3$, $b = -2$, $c = -1$, $d = 1$
 - D. $a = -3$, $b = 2$, $c = 1$, $d = -1$
- 5 The diagram shows the number of penguins, $P(t)$, on an island at time t .



Which equation best represents this graph?

- A. $P(t) = 1500 + 1500e^{-kt}$
- B. $P(t) = 3000 - 1500e^{-kt}$
- C. $P(t) = 3000 + 1500e^{-kt}$
- D. $P(t) = 4500 - 1500e^{-kt}$

- 6 The diagram shows the graph of $y = f(x)$. The equation $f(x) = 0$ has a solution at $x = w$.



Newton's method can be used to give an approximation close to the solution $x = w$.

Which initial approximation, x_1 , will give the second approximation that is closest to the solution $x = w$?

- A. $x_1 = a$
 - B. $x_1 = b$
 - C. $x_1 = c$
 - D. $x_1 = d$
- 7 The velocity of a particle, in metres per second, is given by $v = x^2 + 2$, where x is its displacement in metres from the origin.

What is the acceleration of the particle at $x = 1$?

- A. 2 m s^{-2}
- B. 3 m s^{-2}
- C. 6 m s^{-2}
- D. 12 m s^{-2}

- 8 Six men and six women are to be seated at a round table.

In how many different ways can they be seated if men and women alternate?

- A. $5! 5!$
- B. $5! 6!$
- C. $2! 5! 5!$
- D. $2! 5! 6!$

- 9 Which of the following is a general solution of the equation $\sin 2x = -\frac{1}{2}$?

- A. $x = n\pi + (-1)^n \frac{\pi}{12}$
- B. $x = \frac{n\pi}{2} + (-1)^n \frac{\pi}{12}$
- C. $x = n\pi + (-1)^{n+1} \frac{\pi}{12}$
- D. $x = \frac{n\pi}{2} + (-1)^{n+1} \frac{\pi}{12}$

- 10 A particle is moving in simple harmonic motion. The displacement of the particle is x and its velocity, v , is given by the equation $v^2 = n^2 (2kx - x^2)$, where n and k are constants. The particle is initially at $x = k$.

Which function, in terms of time t , could represent the motion of the particle?

- A. $x = k \cos(nt)$
- B. $x = k \sin(nt) + k$
- C. $x = 2k \cos(nt) - k$
- D. $x = 2k \sin(nt) + k$

Section II

60 marks

Attempt Questions 11–14

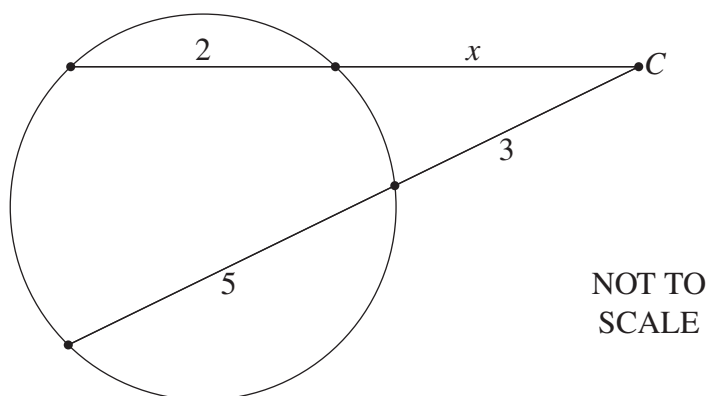
Allow about 1 hour and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet.

- (a) Consider the polynomial $P(x) = x^3 - 2x^2 - 5x + 6$.
- (i) Show that $x = 1$ is a zero of $P(x)$. 1
- (ii) Find the other zeros. 2
- (b) Solve $\log_2 5 + \log_2 (x - 2) = 3$. 2
- (c) Write $\sqrt{3} \sin x + \cos x$ in the form $R \sin(x + \alpha)$ where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$. 2
- (d) Two secants from the point C intersect a circle as shown in the diagram. 2



What is the value of x ?

Question 11 continues on page 7

Question 11 (continued)

(e) Consider the function $f(x) = \frac{1}{4x-1}$.

(i) Find the domain of $f(x)$. **1**

(ii) For what values of x is $f(x) < 1$? **2**

(f) Evaluate $\int_{-3}^0 \frac{x}{\sqrt{1-x}} dx$, using the substitution $u = 1 - x$. **3**

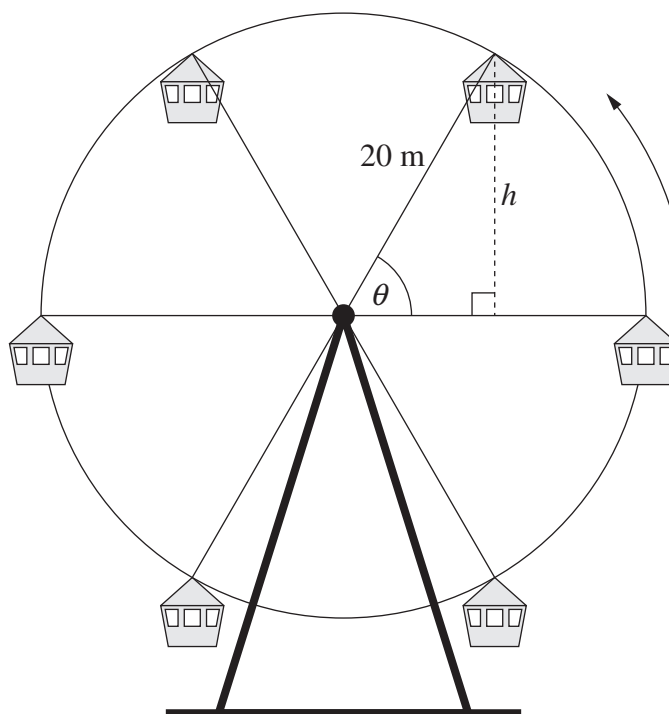
End of Question 11

Question 12 (15 marks) Use the Question 12 Writing Booklet.

(a) Find $\int \cos^2(3x) dx$.

2

- (b) A ferris wheel has a radius of 20 metres and is rotating at a rate of 1.5 radians per minute. The top of a carriage is h metres above the horizontal diameter of the ferris wheel. The angle of elevation of the top of the carriage from the centre of the ferris wheel is θ .



(i) Show that $\frac{dh}{d\theta} = 20 \cos \theta$.

1

- (ii) At what speed is the top of the carriage rising when it is 15 metres higher than the horizontal diameter of the ferris wheel? Give your answer correct to one decimal place.

2

(c) Let $f(x) = \sin^{-1}x + \cos^{-1}x$.

(i) Show that $f'(x) = 0$.

1

(ii) Hence, or otherwise, prove $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$.

1

(iii) Hence, sketch $f(x) = \sin^{-1}x + \cos^{-1}x$.

1

Question 12 continues on page 9

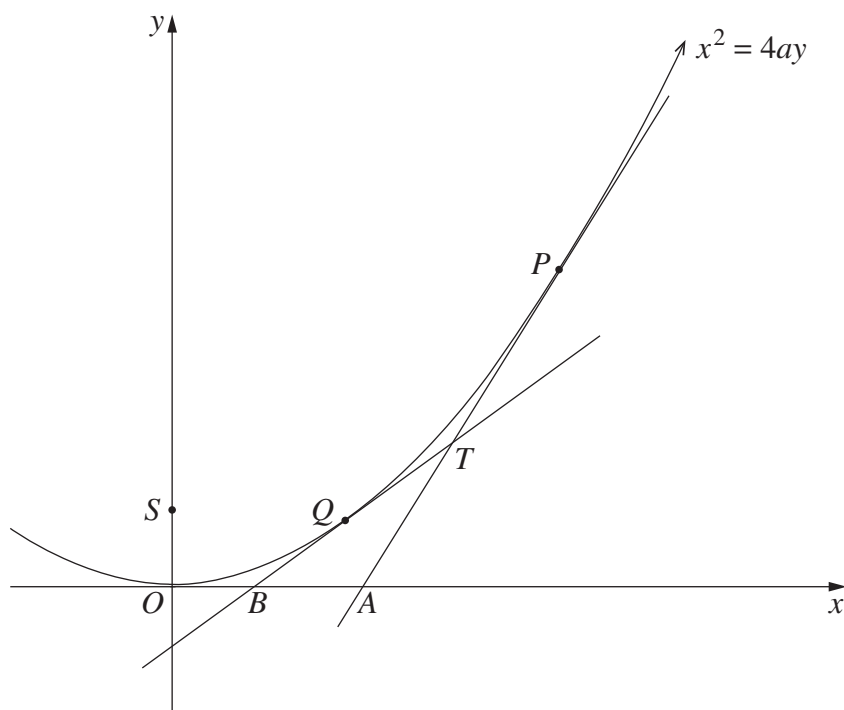
Question 12 (continued)

- (d) A group of 12 people sets off on a trek. The probability that a person finishes the trek within 8 hours is 0.75. 2

Find an expression for the probability that at least 10 people from the group complete the trek within 8 hours.

- (e) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. The focus of the parabola is $S(0, a)$ and the tangents at P and Q intersect at $T(a(p + q), apq)$. (Do NOT prove this.)

The tangents at P and Q meet the x -axis at A and B respectively, as shown.



- (i) Show that $\angle PAS = 90^\circ$. 2
- (ii) Explain why S, B, A, T are concyclic points. 1
- (iii) Show that the diameter of the circle through S, B, A and T has length 2

$$a\sqrt{(p^2 + 1)(q^2 + 1)}.$$

End of Question 12

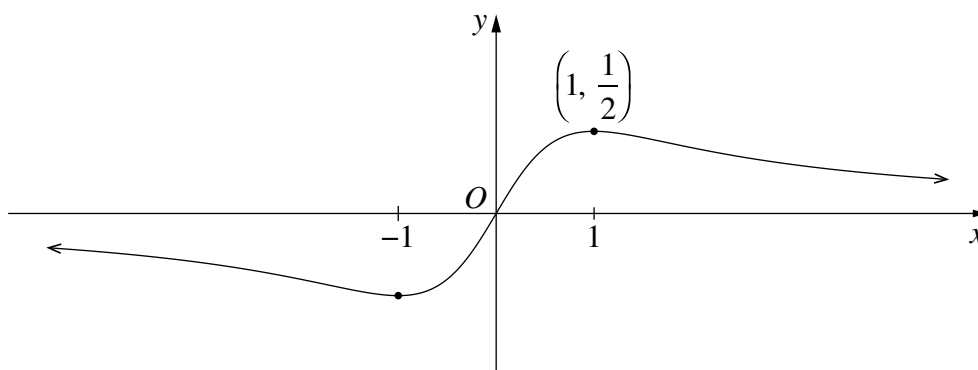
Question 13 (15 marks) Use the Question 13 Writing Booklet.

- (a) Prove by mathematical induction that, for $n \geq 1$,

3

$$2 - 6 + 18 - 54 + \cdots + 2(-3)^{n-1} = \frac{1 - (-3)^n}{2}.$$

- (b) The diagram shows the graph $y = \frac{x}{x^2 + 1}$, for all real x .



Consider the function $f(x) = \frac{x}{x^2 + 1}$, for $x \geq 1$.

The function $f(x)$ has an inverse. (Do NOT prove this.)

- (i) State the domain and range of $f^{-1}(x)$.

2

- (ii) Sketch the graph $y = f^{-1}(x)$.

1

- (iii) Find an expression for $f^{-1}(x)$.

3

Question 13 continues on page 11

Question 13 (continued)

- (c) An object is projected from the origin with an initial velocity of V at an angle θ to the horizontal. The equations of motion of the object are

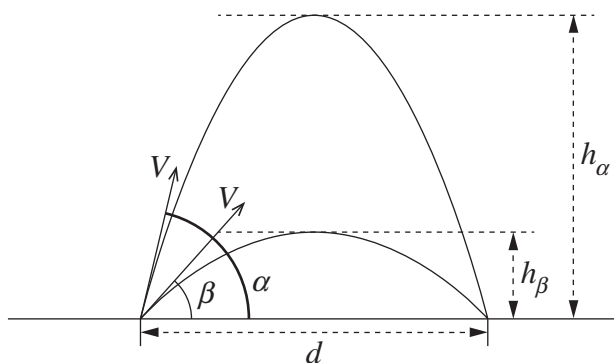
$$x(t) = Vt \cos \theta$$

$$y(t) = Vt \sin \theta - \frac{gt^2}{2}. \quad (\text{Do NOT prove these.})$$

- (i) Show that when the object is projected at an angle θ , the horizontal range is $\frac{V^2}{g} \sin 2\theta$. 2
- (ii) Show that when the object is projected at an angle $\frac{\pi}{2} - \theta$, the horizontal range is also $\frac{V^2}{g} \sin 2\theta$. 1
- (iii) The object is projected with initial velocity V to reach a horizontal distance d , which is less than the maximum possible horizontal range. There are two angles at which the object can be projected in order to travel that horizontal distance before landing. Let these angles be α and β , where $\beta = \frac{\pi}{2} - \alpha$. 3

Let h_α be the maximum height reached by the object when projected at the angle α to the horizontal.

Let h_β be the maximum height reached by the object when projected at the angle β to the horizontal.

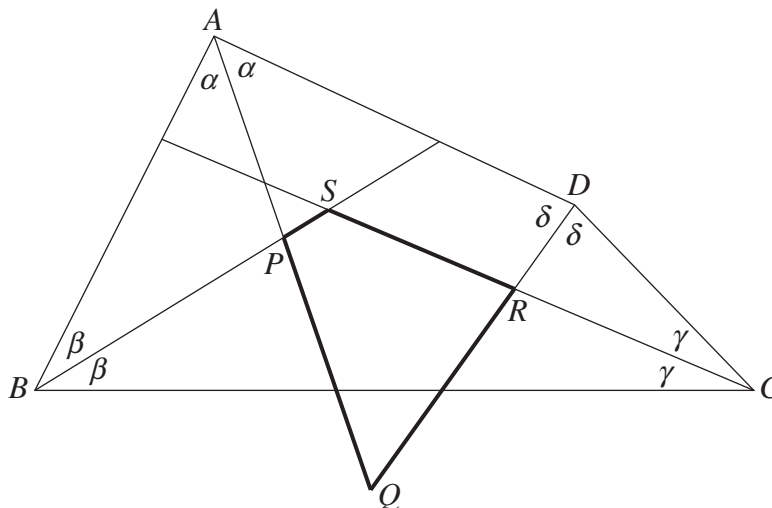


Show that the average of the two heights, $\frac{h_\alpha + h_\beta}{2}$, depends only on V and g .

End of Question 13

Question 14 (15 marks) Use the Question 14 Writing Booklet.

- (a) The diagram shows quadrilateral $ABCD$ and the bisectors of the angles at A , B , C and D . The bisectors at A and B intersect at the point P . The bisectors at A and D meet at Q . The bisectors at C and D meet at R . The bisectors at B and C meet at S . **3**



Copy or trace the diagram into your writing booklet.

Show that $PQRS$ is a cyclic quadrilateral.

- (b) (i) By considering the expansions of $(1 + (1 + x))^n$ and $(2 + x)^n$, show that **3**

$$\binom{n}{r}\binom{r}{r} + \binom{n}{r+1}\binom{r+1}{r} + \binom{n}{r+2}\binom{r+2}{r} + \cdots + \binom{n}{n}\binom{n}{r} = \binom{n}{r}2^{n-r}.$$

- (ii) There are 23 people who have applied to be selected for a committee of 4 people. **2**

The selection process starts with Selector A choosing a group of at least 4 people from the 23 people who applied.

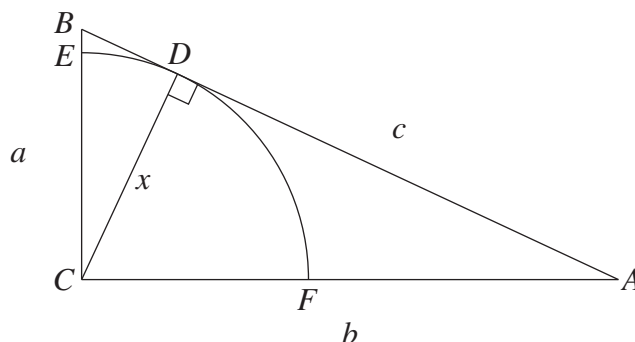
Selector B then chooses the 4 people to be on the committee from the group Selector A has chosen.

In how many ways could this selection process be carried out?

Question 14 continues on page 13

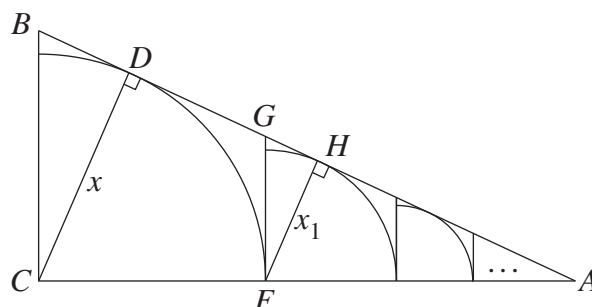
Question 14 (continued)

- (c) In triangle ABC , BC is perpendicular to AC . Side BC has length a , side AC has length b and side AB has length c . A quadrant of a circle of radius x , centred at C , is constructed. The arc meets side BC at E . It touches the side AB at D , and meets side AC at F . The interval CD is perpendicular to AB .



- (i) Show that $\triangle ABC$ and $\triangle ACD$ are similar. 1
- (ii) Show that $x = \frac{ab}{c}$. 1

From F , a line perpendicular to AC is drawn to meet AB at G , forming the right-angled triangle GFA . A new quadrant is constructed in triangle GFA touching side AB at H . The process is then repeated indefinitely.



- (iii) Show that the limiting sum of the areas of all the quadrants is $\frac{\pi ab^2}{4(2c - a)}$. 4
- (iv) Hence, or otherwise, show that $\frac{\pi}{2} < \frac{2c - a}{b}$. 1

End of paper

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2018 HIGHER SCHOOL CERTIFICATE EXAMINATION

REFERENCE SHEET

- Mathematics –
- Mathematics Extension 1 –
- Mathematics Extension 2 –

Mathematics

Factorisation

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Angle sum of a polygon

$$S = (n - 2) \times 180^\circ$$

Equation of a circle

$$(x - h)^2 + (y - k)^2 = r^2$$

Trigonometric ratios and identities

$$\sin \theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

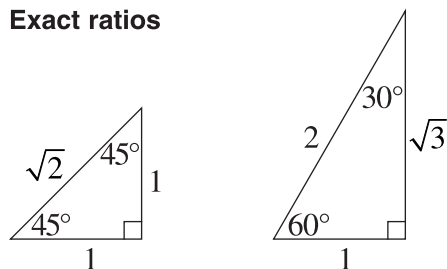
$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Exact ratios



Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Cosine rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Area of a triangle

$$\text{Area} = \frac{1}{2} ab \sin C$$

Distance between two points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Perpendicular distance of a point from a line

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Slope (gradient) of a line

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-gradient form of the equation of a line

$$y - y_1 = m(x - x_1)$$

n th term of an arithmetic series

$$T_n = a + (n - 1)d$$

Sum to n terms of an arithmetic series

$$S_n = \frac{n}{2} [2a + (n - 1)d] \quad \text{or} \quad S_n = \frac{n}{2} (a + l)$$

n th term of a geometric series

$$T_n = ar^{n-1}$$

Sum to n terms of a geometric series

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{or} \quad S_n = \frac{a(1 - r^n)}{1 - r}$$

Limiting sum of a geometric series

$$S = \frac{a}{1 - r}$$

Compound interest

$$A_n = P \left(1 + \frac{r}{100} \right)^n$$

Mathematics (continued)

Differentiation from first principles

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Derivatives

If $y = x^n$, then $\frac{dy}{dx} = nx^{n-1}$

If $y = uv$, then $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$

If $y = \frac{u}{v}$, then $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

If $y = F(u)$, then $\frac{dy}{dx} = F'(u) \frac{du}{dx}$

If $y = e^{f(x)}$, then $\frac{dy}{dx} = f'(x)e^{f(x)}$

If $y = \log_e f(x) = \ln f(x)$, then $\frac{dy}{dx} = \frac{f'(x)}{f(x)}$

If $y = \sin f(x)$, then $\frac{dy}{dx} = f'(x) \cos f(x)$

If $y = \cos f(x)$, then $\frac{dy}{dx} = -f'(x) \sin f(x)$

If $y = \tan f(x)$, then $\frac{dy}{dx} = f'(x) \sec^2 f(x)$

Solution of a quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum and product of roots of a quadratic equation

$$\alpha + \beta = -\frac{b}{a} \quad \alpha\beta = \frac{c}{a}$$

Equation of a parabola

$$(x-h)^2 = \pm 4a(y-k)$$

Integrals

$$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$$

$$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

$$\int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + C$$

$$\int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + C$$

$$\int \sec^2(ax+b) dx = \frac{1}{a} \tan(ax+b) + C$$

Trapezoidal rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)]$$

Simpson's rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Logarithms – change of base

$$\log_a x = \frac{\log_b x}{\log_b a}$$

Angle measure

$$180^\circ = \pi \text{ radians}$$

Length of an arc

$$l = r\theta$$

Area of a sector

$$\text{Area} = \frac{1}{2} r^2 \theta$$

Mathematics Extension 1

Angle sum identities

$$\sin(\theta + \phi) = \sin\theta \cos\phi + \cos\theta \sin\phi$$

$$\cos(\theta + \phi) = \cos\theta \cos\phi - \sin\theta \sin\phi$$

$$\tan(\theta + \phi) = \frac{\tan\theta + \tan\phi}{1 - \tan\theta \tan\phi}$$

t formulae

If $t = \tan \frac{\theta}{2}$, then

$$\sin\theta = \frac{2t}{1+t^2}$$

$$\cos\theta = \frac{1-t^2}{1+t^2}$$

$$\tan\theta = \frac{2t}{1-t^2}$$

General solution of trigonometric equations

$$\sin\theta = a, \quad \theta = n\pi + (-1)^n \sin^{-1}a$$

$$\cos\theta = a, \quad \theta = 2n\pi \pm \cos^{-1}a$$

$$\tan\theta = a, \quad \theta = n\pi + \tan^{-1}a$$

Division of an interval in a given ratio

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

Parametric representation of a parabola

For $x^2 = 4ay$,

$$x = 2at, \quad y = at^2$$

At $(2at, at^2)$,

$$\text{tangent: } y = tx - at^2$$

$$\text{normal: } x + ty = at^3 + 2at$$

At (x_1, y_1) ,

$$\text{tangent: } xx_1 = 2a(y + y_1)$$

$$\text{normal: } y - y_1 = -\frac{2a}{x_1}(x - x_1)$$

Chord of contact from (x_0, y_0) : $xx_0 = 2a(y + y_0)$

Acceleration

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

Simple harmonic motion

$$x = b + a \cos(nt + \alpha)$$

$$\ddot{x} = -n^2(x - b)$$

Further integrals

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

Sum and product of roots of a cubic equation

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

Estimation of roots of a polynomial equation

Newton's method

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

Binomial theorem

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

2018 HSC Mathematics Extension 1 Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	B
2	A
3	A
4	D
5	A
6	C
7	C
8	B
9	D
10	B

Section II

Question 11 (a) (i)

Criteria	Marks
• Provides correct proof	

Sample answer:

$$\begin{aligned}
 P(1) &= 1^3 - 2(1)^2 - 5(1) + 6 \\
 &= 1 - 2 - 5 + 6 \\
 &= 0
 \end{aligned}$$

$\therefore x = 1$ is a zero of $P(x)$

Question 11 (a) (ii)

Criteria	Marks
• Finds correct solution	2
• Correctly divides by $x - 1$, or equivalent merit	1

Sample answer:

$$\begin{array}{r}
 x^2 - x - 6 \\
 x - 1 \overline{) x^3 - 2x^2 - 5x + 6} \\
 \underline{x^3 - x^2} \\
 -x^2 - 5x \\
 \underline{-x^2 + x} \\
 -6x + 6 \\
 \underline{-6x + 6} \\
 0
 \end{array}$$

$$\begin{aligned}
 \therefore P(x) &= (x - 1)(x^2 - x - 6) \\
 &= (x - 1)(x + 2)(x - 3)
 \end{aligned}$$

other zeros are $-2, 3$

Question 11 (b)

Criteria	Marks
• Provides correct solution	2
• Obtains correct linear equation, or equivalent merit	1

Sample answer:

$$\log_2 5 + \log_2 (x - 2) = 3$$

$$\log_2 (5(x - 2)) = 3$$

$$5(x - 2) = 2^3$$

$$5x - 10 = 8$$

$$5x = 18$$

$$x = \frac{18}{5} \text{ or } 3\frac{3}{5}$$

Question 11 (c)

Criteria	Marks
• Provides correct solution	2
• Correctly finds R , or equivalent merit	1

Sample answer:

$$\sqrt{3} \sin x + \cos x = R \sin(x + \alpha), \quad R > 0, \quad 0 \leq \alpha \leq \frac{\pi}{2}$$

$$= R \sin x \cos \alpha + R \cos x \sin \alpha$$

$$\therefore R \cos \alpha = \sqrt{3} \quad \textcircled{1}$$

$$R \sin \alpha = 1 \quad \textcircled{2}$$

$$\textcircled{2} \div \textcircled{1} \quad \tan \alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = \frac{\pi}{6}$$

$$\textcircled{2}^2 + \textcircled{1}^2 \quad R^2 (\sin^2 \alpha + \cos^2 \alpha) = 1 + 3$$

$$R^2 = 4$$

$$R = 2 \quad (R > 0)$$

$$\therefore \sqrt{3} \sin x + \cos x = 2 \sin \left(x + \frac{\pi}{6} \right)$$

Question 11 (d)

Criteria	Marks
• Provides correct solution	2
• Attempts to use secant theorem resulting in a quadratic equation, or equivalent merit	1

Sample answer:

$$x(x+2) = 3(3+5)$$

$$x^2 + 2x = 24$$

$$x^2 + 2x - 24 = 0$$

$$(x+6)(x-4) = 0$$

$$x = -6 \text{ or } 4$$

But $x > 0 \therefore x = 4$

Question 11 (e) (i)

Criteria	Marks
• Provides correct answer	1

Sample answer:

Domain $4x - 1 \neq 0$

$$4x \neq 1$$

$$x \neq \frac{1}{4}$$

Domain: all real $x \neq \frac{1}{4}$

Question 11 (e) (ii)

Criteria	Marks
• Provides correct solution	2
• Successfully deals with the denominator, or equivalent merit	1

Sample answer:

$$f(x) < 1$$

$$\frac{1}{4x-1} < 1 \quad x \neq \frac{1}{4}$$

$$(4x-1)^2 \times \frac{1}{4x-1} < (4x-1)^2 \times 1$$

$$4x-1 < (4x-1)^2$$

$$0 < (4x-1)^2 - (4x-1)$$

$$0 < (4x-1)(4x-1-1)$$

$$0 < (4x-1)(4x-2)$$

$$0 < 2(4x-1)(2x-1)$$

$$\therefore x < \frac{1}{4}, x > \frac{1}{2}$$

Question 11 (f)

Criteria	Marks
• Provides correct solution	3
• Obtains correct integral in terms of u , including the limits, or equivalent merit	2
• Attempts to use the substitution, or equivalent merit	1

Sample answer:

$$I = \int_{-3}^0 \frac{x}{\sqrt{1-x}} dx$$

$$= \int_4^1 \frac{1-u}{\sqrt{u}} (-du)$$

$$= \int_1^4 u^{-\frac{1}{2}} - u^{\frac{1}{2}} du$$

$$= \left[2u^{\frac{1}{2}} - \frac{2}{3}u^{\frac{3}{2}} \right]_1^4$$

$$= 2(2) - \frac{2}{3}(8) - \left(2(1) - \frac{2}{3}(1) \right)$$

$$= 4 - 5\frac{1}{3} - 1\frac{1}{3}$$

$$= -2\frac{2}{3}$$

$$\text{Let } u = 1 - x$$

$$du = -dx$$

$$x = -3 \rightarrow u = 4$$

$$x = 0 \rightarrow u = 1$$

Question 12 (a)

Criteria	Marks
• Provides correct solution	2
• Rewrites the integrand using double angle result, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 & \int \cos^2(3x) dx \\
 &= \frac{1}{2} \int \cos(6x) + 1 dx \quad \text{since } \cos^2 \theta = \frac{1}{2}(\cos 2\theta + 1) \\
 &= \frac{1}{2} \left(\frac{1}{6} \sin 6x + x \right) + c \\
 &= \frac{1}{12} \sin 6x + \frac{x}{2} + c
 \end{aligned}$$

Question 12 (b) (i)

Criteria	Marks
• Provides correct proof	1

Sample answer:

$$\begin{aligned}
 \frac{h}{20} &= \sin \theta \\
 h &= 20 \sin \theta \\
 \frac{dh}{d\theta} &= 20 \cos \theta
 \end{aligned}$$

Question 12 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Uses related rates to find an expression for $\frac{dh}{dt}$, or equivalent merit	1

Sample answer:

When $h = 15$ $\theta = ?$

$$\sin \theta = \frac{15}{20} = \frac{3}{4}$$

$$\theta = \sin^{-1}\left(\frac{3}{4}\right)$$

we want $\frac{dh}{dt} = \frac{dh}{d\theta} \times \frac{d\theta}{dt}$

$$\frac{dh}{d\theta} \text{ when } h = 15 \quad \frac{dh}{d\theta} = 20 \cos\left(\sin^{-1}\left(\frac{3}{4}\right)\right)$$

and $\frac{d\theta}{dt} = 1.5$ radians/minute (given)

$$\begin{aligned} \text{so } \frac{dh}{dt} &= 20 \cos\left(\sin^{-1}\left(\frac{3}{4}\right)\right) \times 1.5 \\ &= 30 \cos\left(\sin^{-1}\left(\frac{3}{4}\right)\right) \\ &= 19.8 \end{aligned}$$

Question 12 (c) (i)

Criteria	Marks
• Provides correct proof	1

Sample answer:

$$f(x) = \sin^{-1} x + \cos^{-1} x$$

$$\begin{aligned} f'(x) &= \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}} \\ &= 0 \end{aligned}$$

Question 12 (c) (ii)

Criteria	Marks
• Provides correct proof	1

Sample answer:

$$f(x) = \sin^{-1} x + \cos^{-1} x$$

Since $f'(x) = 0$

$f(x)$ must have a constant value on its domain, so only need to find $f(a)$ for some a .

When $x = 0$

$$f(0) = \sin^{-1} 0 + \cos^{-1} 0$$

$$= 0 + \frac{\pi}{2} = \frac{\pi}{2}$$

or

$$\text{Let } m = \sin^{-1} x$$

$$\sin m = x$$

$$\therefore \cos\left(\frac{\pi}{2} - m\right) = x$$

$$\text{Let } \frac{\pi}{2} - m = \cos^{-1} x$$

$$\begin{aligned} \therefore \sin^{-1} x + \cos^{-1} x &= m + \frac{\pi}{2} - m \\ &= \frac{\pi}{2} \end{aligned}$$

Question 12 (c) (iii)

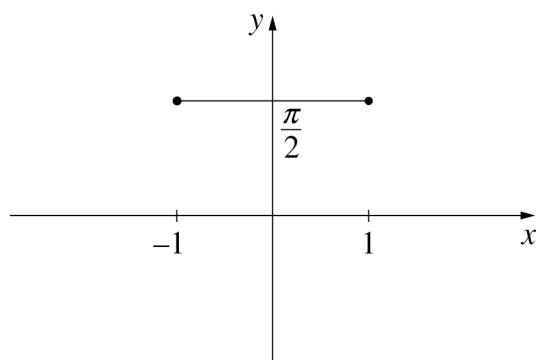
Criteria	Marks
• Provides correct sketch	1

Sample answer:

Sketch $f(x) = \sin^{-1} x + \cos^{-1} x$

domain $\sin^{-1} x$ is $-1 \leq x \leq 1$
 $\cos^{-1} x$ is $-1 \leq x \leq 1$

so domain $f(x)$ is $-1 \leq x \leq 1$



Question 12 (d)

Criteria	Marks
• Provides correct solution	2
• Obtains correct probability for 10 people completing the trek, or equivalent merit	1

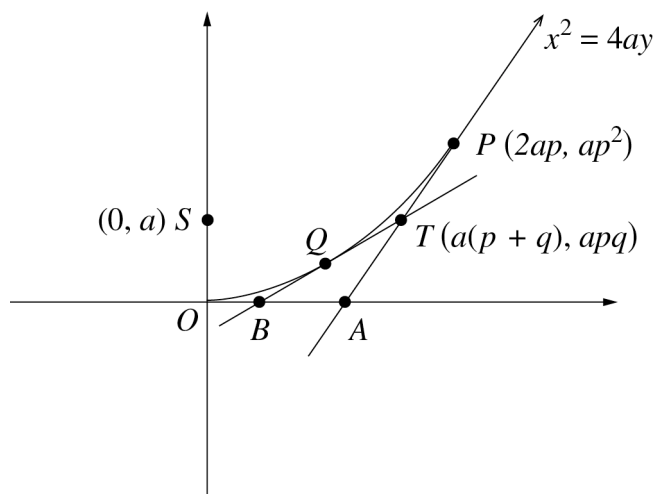
Sample answer:

$$\binom{12}{10}(0.75)^{10}(0.25)^2 + \binom{12}{11}(0.75)^{11}(0.25) + \binom{12}{12}(0.75)^{12}$$

Question 12 (e) (i)

Criteria	Marks
• Provides correct solution	2
• Finds the gradient of AS , or equivalent merit	1

Sample answer:



Tangent at P $y = px - ap^2$

so the coordinates of A are $(ap, 0)$

$$\begin{aligned}
 \text{the line joining } A \text{ and } S \text{ has slope} &= \frac{a - 0}{0 - ap} \\
 &= -\frac{a}{ap} \\
 &= -\frac{1}{p}
 \end{aligned}$$

$$\begin{aligned}
 \text{slope of tangent} \times \text{slope of } AS &= -\frac{1}{p} \times p \\
 &= -1
 \end{aligned}$$

lines are perpendicular

so $\angle PAS = 90^\circ$

Question 12 (e) (ii)

Criteria	Marks
• Provides correct explanation	1

Sample answer:

$$\angle PAS = 90^\circ \text{ which means } \angle TAS = 90^\circ$$

similarly we could show that

$$\angle QBS = 90^\circ \text{ so } \angle TBS = 90^\circ$$

So we have equal angles subtended on the same chord (on the same arc).

So the points A , B , S and T are concyclic points.

Question 12 (e) (iii)

Criteria	Marks
• Provides correct solution	2
• Correctly substitutes S and T into the distance formula, or equivalent merit	1

Sample answer:

ST is the diameter of the circle since $\angle TAS = \angle TBS = 90^\circ$

so we want the length of interval ST

coordinates of $S = (0, a)$ coordinates $T = (a(p + q), apq)$

$$\begin{aligned}
 \text{distance} &= \sqrt{(a(p + q) - 0)^2 + (apq - a)^2} \\
 &= \sqrt{a^2(p + q)^2 + a^2(pq - 1)^2} \\
 &= \sqrt{a^2(p^2 + 2pq + q^2 + p^2q^2 - 2pq + 1)} \\
 &= a\sqrt{p^2 + q^2 + 1 + p^2q^2} \\
 &= a\sqrt{(p^2 + 1)(q^2 + 1)}
 \end{aligned}$$

Question 13 (a)

Criteria	Marks
• Provides correct proof	3
• Correctly sets up S_{k+1} as the sum of S_k and T_{k+1} , or equivalent merit	2
• Proves true for $n = 1$, or equivalent merit	1

Sample answer:

$$\text{RTP: } 2 - 6 + 18 - 54 + \dots + 2(-3)^{n-1} = \frac{1 - (-3)^n}{2}$$

Show true for $n = 1$:

$$\begin{aligned} \text{LHS} &= 2(-3)^{1-1} \\ &= 2 \times 1 \\ &= 2 \end{aligned} \qquad \begin{aligned} \text{RHS} &= \frac{1 - (-3)^1}{2} \\ &= \frac{1 + 3}{2} \\ &= 2 \end{aligned}$$

LHS = RHS $\therefore$ true for $n = 1$

Assume true for $n = k$:

$$2 - 6 + 18 - 54 + \dots + 2(-3)^{k-1} = \frac{1 - (-3)^k}{2}$$

If $n = k$ is true, show true for $n = k + 1$.

$$\begin{aligned} \text{LHS} &= 2 - 6 + 18 - 54 + \dots + 2(-3)^{k-1} + 2(-3)^k \\ &= \frac{1 - (-3)^k}{2} + 2(-3)^k \\ &= \frac{1 - (-3)^k + 4(-3)^k}{2} \\ &= \frac{1 - (-3)(-3)^k}{2} \\ &= \frac{1 - (-3)^{k+1}}{2} \end{aligned}$$

$\therefore$ true for $n = k + 1$.

Hence, by the process of mathematical induction $2 - 6 + 18 - 54 + \dots + 2(-3)^{n-1} = \frac{1 - (-3)^n}{2}$.

Question 13 (b) (i)

Criteria	Marks
• Provides correct solution	2
• States correct domain, or equivalent merit	1

Sample answer:

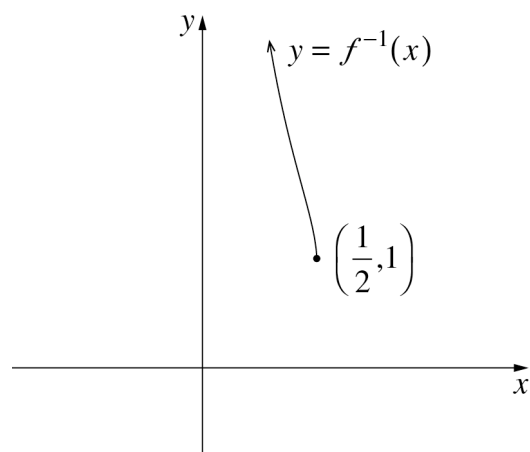
Domain: all real x , $0 < x \leq \frac{1}{2}$

Range: all real, $y \geq 1$

Question 13 (b) (ii)

Criteria	Marks
• Provides correct sketch	1

Sample answer:



Question 13 (b) (iii)

Criteria	Marks
• Provides correct solution	3
• Correctly solves the resulting quadratic equation, or equivalent merit	2
• Correctly interchanges x and y , or equivalent merit	1

Sample answer:

Let $y = \frac{x}{x^2 + 1}$

$$x^2y + y = x$$

$$x^2y - x + y = 0$$

$$x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4 \times y \times y}}{2y}$$

$$= \frac{1 \pm \sqrt{1 - 4y^2}}{2y}$$

when $x = 2, y = \frac{2}{5}$

$$\therefore x = \frac{1 + \sqrt{1 - 4y^2}}{2y}$$

$$\therefore f^{-1}(x) = \frac{1 + \sqrt{1 - 4x^2}}{2x}$$

Question 13 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Attempts to solve $y = 0$, or equivalent merit	1

Sample answer:

The object hits the ground, after $t = 0$ when

$$\begin{aligned}
 0 &= y(t) \\
 &= Vt \sin \theta - \frac{gt^2}{2} \\
 &= t \left(V \sin \theta - \frac{gt}{2} \right)
 \end{aligned}$$

As $t > 0$ we have

$$\begin{aligned}
 t &= \frac{2V \sin \theta}{g} \\
 x &= Vt \cos \theta \\
 &= \frac{2V^2 \sin \theta \cos \theta}{g} \\
 &= \frac{V^2}{g} \sin 2\theta
 \end{aligned}$$

Thus the range, at angle θ , is $\frac{V^2}{g} \sin 2\theta$.

Question 13 (c) (ii)

Criteria	Marks
• Provides correct proof	1

Sample answer:

$$\begin{aligned}
 \text{Horizontal range} &= \frac{V^2}{g} \sin 2 \left(\frac{\pi}{2} - \theta \right) \\
 &= \frac{V^2}{g} \sin(\pi - 2\theta) \\
 &= \frac{V^2}{g} \sin 2\theta
 \end{aligned}$$

Question 13 (c) (iii)

Criteria	Marks
• Provides correct solution	3
• Finds h_{α} or equivalent merit	2
• Finds time when maximum height is reached, or equivalent merit	1

Sample answer:

The maximum height, for angle θ , occurs when $\dot{y}(t) = 0$, that is

$$\begin{aligned}
 0 &= \dot{y}(t) \\
 &= V \sin \theta - gt \\
 t &= \frac{V \sin \theta}{g} \\
 y\left(\frac{V \sin \theta}{g}\right) &= \frac{V^2 \sin^2 \theta}{g} - \frac{V^2 \sin^2 \theta}{2g} \\
 &= \frac{V^2 \sin^2 \theta}{2g}
 \end{aligned}$$

Thus $h_{\alpha} = \frac{V^2 \sin^2 \alpha}{2g}$ and $h_{\beta} = \frac{V^2 \sin^2 \beta}{2g}$, but

$$\sin \beta = \sin\left(\frac{\pi}{2} - \alpha\right) = \cos \alpha$$

Thus

$$\begin{aligned}
 h_{\alpha} + h_{\beta} &= \frac{V^2 \sin^2 \alpha}{2g} + \frac{V^2 \sin^2 \beta}{2g} \\
 &= \frac{V^2 \sin^2 \alpha}{2g} + \frac{V^2 \cos^2 \alpha}{2g} \\
 &= \frac{V^2 (\sin^2 \alpha + \cos^2 \alpha)}{2g} \\
 &= \frac{V^2}{2g} \\
 \frac{h_{\alpha} + h_{\beta}}{2} &= \frac{V^2}{4g}
 \end{aligned}$$

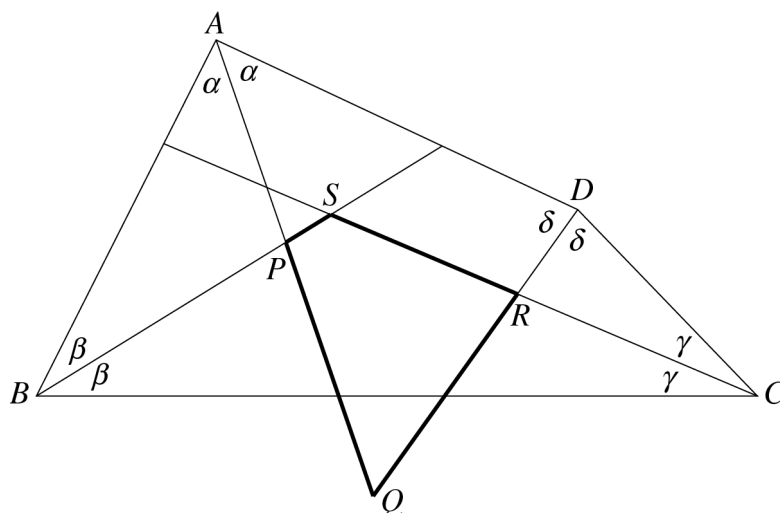
which depends only on V and g .

In fact, this is the height attained by the object when projected at $\frac{\pi}{4}$ to attain its maximum range.

Question 14 (a)

Criteria	Marks
• Provides correct proof	3
• Establishes a relationship between the angles of the quadrilateral $PQRS$ and the bisected angles, or equivalent merit	2
• Relates the sum of the bisected angles to the angle sum of the quadrilateral $ABCD$, or equivalent merit	1

Sample answer:



$$\text{Let } \alpha = \frac{1}{2}\angle A, \quad \beta = \frac{1}{2}\angle B$$

$$\gamma = \frac{1}{2}\angle C \quad \text{and} \quad \delta = \frac{1}{2}\angle D$$

as shown

$$\text{Then } 2\alpha + 2\beta + 2\gamma + 2\delta = 360^\circ \quad (\angle \text{ sum quadrilateral})$$

$$\therefore \alpha + \beta + \gamma + \delta = 180^\circ$$

$$\angle PSR = \angle BSC = 180^\circ - (\beta + \gamma) \quad (\angle \text{ sum of } \triangle BSC)$$

$$\angle PQR = \angle AQD = 180^\circ - (\alpha + \delta) \quad (\angle \text{ sum of } \triangle AQD)$$

$$\begin{aligned} \therefore \angle PSR + \angle PQR &= 2 \times 180^\circ - (\alpha + \beta + \gamma + \delta) \\ &= 2 \times 180^\circ - 180^\circ \\ &= 180^\circ \end{aligned}$$

$\therefore PQRS$ is cyclic (opposite angles are supplementary)

Question 14 (b) (i)

Criteria	Marks
• Provides correct solution	3
• Obtains correct expansion of $(1 + (1 + x)^n)$ in terms of $(1 + x)^k$, or equivalent merit	2
• Correctly applies the binomial theorem to one binomial expression to get the coefficient of x^r , or equivalent merit	1

Sample answer:

Consider the coefficients of x^r on both sides of the equation

$$(2 + x)^n = (1 + (1 + x))^n$$

$$\text{Then in } x^r \text{ on LHS} = \binom{n}{r} x^r 2^{n-r}$$

$$\therefore \text{coefficient of } x^r = \binom{n}{r} 2^{n-r}$$

$$\text{RHS} = \binom{n}{0} + \binom{n}{1}(1 + x)^1 + \binom{n}{2}(1 + x)^2 + \dots + \binom{n}{r}(1 + x)^r + \binom{n}{r+1}(1 + x)^{r+1} + \dots + \binom{n}{n}(1 + x)^n$$

Coefficient of x^r = sum of coefficients of x^r for each term

$$= \binom{n}{r} \binom{r}{r} + \binom{n}{r+1} \binom{r+1}{r} + \binom{n}{r+2} \binom{r+2}{r} + \dots + \binom{n}{n} \binom{n}{r}$$

Equating these gives the result.

Question 14 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Attempts to use the formula in part (i), or equivalent merit	1

Sample answer:

Selector *A* could choose a group of 4, 5, 6 ... or 23 people. Then selector *B* chooses 4 from this group.

Hence number of ways

$$\begin{aligned}
 &= \binom{23}{4} \binom{4}{4} + \binom{23}{5} \binom{5}{4} + \binom{23}{6} \binom{6}{4} + \dots + \binom{23}{23} \binom{23}{4} \\
 &= \binom{23}{4} 2^{23-4} \\
 &= \binom{23}{4} (2^{19}) \\
 &= 4\,642\,570\,240
 \end{aligned}$$

Question 14 (c) (i)

Criteria	Marks
• Provides correct proof	1

Sample answer:

In $\triangle ABC$ and $\triangle ACD$

$\angle ACB = \angle ADC$ (CD perpendicular to AB , $\triangle ABC$ right-angled at C)

$\angle BAC = \angle CAD$ (common)

$\therefore \triangle ABC \parallel \triangle ACD$ (equiangular)

Question 14 (c) (ii)

Criteria	Marks
• Provides correct proof	1

Sample answer:

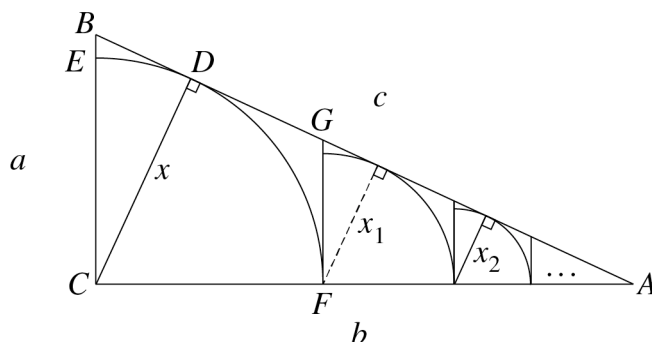
$$\frac{x}{a} = \frac{b}{c} \quad (\text{corresponding sides of similar triangles})$$

$$\therefore x = \frac{ab}{c}$$

Question 14 (c) (iii)

Criteria	Marks
• Provides correct solution	4
• Correctly establishes a relationship between successive radii or area, or equivalent merit	3
• Attempts to use similarity to find a relationship between successive radii or area, or equivalent merit	2
• Finds an expression for the area of a quadrant in terms of a , b and c , or equivalent merit	1

Sample answer:



Since $\triangle ABC$ and $\triangle AGF$ are right-angled and angle A is common, then $\triangle ABC \parallel \triangle AGF$

$$\frac{GF}{a} = \frac{b-x}{b}$$

$$\frac{GF}{a} = \frac{1}{b} \left(b - \frac{ab}{c} \right)$$

$$\frac{GF}{a} = \frac{b}{b} \left(1 - \frac{a}{c} \right)$$

It follows then

$$\frac{x_1}{x} = 1 - \frac{a}{c}$$

$$x_1 = \left(1 - \frac{a}{c} \right) x$$

$$x_1 = (1 - \sin A)x$$

The ratio of the lengths of the radii are independent of a , b and c .

Hence, successive radii will be in the ratio $(1 - \sin A)$.

$$\text{Hence, } x_2 = \left(1 - \frac{a}{c} \right)^2 x \text{ etc}$$

Area of quadrants

$$A = \frac{1}{4}\pi(x^2 + x_1^2 + x_2^2 + \dots)$$

$$A = \frac{1}{4}\pi\left(x^2 + \left(1 - \frac{a}{c}\right)^2 x^2 + \left(1 - \frac{a}{c}\right)^4 x^2 + \dots\right)$$

$$A = \frac{1}{4}\pi x^2 \left(1 + \left(1 - \frac{a}{c}\right)^2 + \left(1 - \frac{a}{c}\right)^4 + \dots\right)$$

For the series

$$1 + \left(1 - \frac{a}{c}\right)^2 + \left(1 - \frac{a}{c}\right)^4 + \dots$$

$$a = 1, \quad r = \left(1 - \frac{a}{c}\right)^2$$

$$S_{\infty} = \frac{a}{1-r}$$

$$S_{\infty} = \frac{1}{1 - \left(1 - \frac{a}{c}\right)^2}$$

$$S_{\infty} = \frac{c^2}{c^2 - (c-a)^2}$$

$$\begin{aligned} S_{\infty} &= \frac{c^2}{c^2 - c^2 + 2ac - a^2} \\ &= \frac{c^2}{a(2c-a)} \end{aligned}$$

$$\therefore A = \frac{1}{4}\pi x^2 \times \frac{c^2}{a(2c-a)}$$

$$A = \frac{\pi}{4} \left(\frac{ab}{c}\right)^2 \times \frac{c^2}{a(2c-a)}$$

$$A = \frac{\pi}{4} \times \frac{ab^2}{(2c-a)}$$

$$A = \frac{\pi ab^2}{4(2c-a)}$$

Question 14 (c) (iv)

Criteria	Marks
• Provides correct solution	1

Sample answer:

Area of quadrants < area of $\triangle ABC$

$$\frac{\pi ab^2}{4(2c-a)} < \frac{ab}{2}$$

$$\frac{\pi}{4} < \frac{ab}{2} \times \frac{(2c-a)}{ab^2}$$

$$\frac{\pi}{4} < \frac{2c-a}{2b}$$

$$\frac{\pi}{2} < \frac{2c-a}{b}$$

2018 HSC Mathematics Extension 1 Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	16.3	PE3
2	1	6.6	PE6
3	1	5.7, 13.4	HE7
4	1	16.1	PE3
5	1	14.2E	HE3
6	1	16.4	HE7
7	1	14.3E	HE5
8	1	18.1	PE3
9	1	15.2	HE7
10	1	14.4E	HE5

Section II

Question	Marks	Content	Syllabus outcomes
11 (a) (i)	1	16.2	PE3
11 (a) (ii)	2	16.2	PE3
11 (b)	2	12.2	H3, HE7
11 (c)	2	5.9E, 13.1	HE7
11 (d)	2	2.10	PE3
11 (e) (i)	1	4.1	PE2
11 (e) (ii)	2	1.4E	PE3
11 (f)	3	11.5E	HE6
12 (a)	2	13.6E	HE7
12 (b) (i)	1	14.1E, 14.4E	HE3, HE5
12 (b) (ii)	2	14.1E, 14.4E	HE3, HE5
12 (c) (i)	1	15.5	HE4
12 (c) (ii)	1	15.2, 15.4	HE4
12 (c) (iii)	1	15.3	HE4
12 (d)	2	18.2	HE3
12 (e) (i)	2	9.6	PE3
12 (e) (ii)	1	2.9	PE3
12 (e) (iii)	2	2.9, 6.5	PE3
13 (a)	3	7.4	HE2
13 (b) (i)	2	15.1	HE4
13 (b) (ii)	1	15.1E	HE4
13 (b) (iii)	3	15.1E	HE4
13 (c) (i)	2	14.3E	HE3
13 (c) (ii)	1	14.3E	HE3

Question	Marks	Content	Syllabus outcomes
13 (c) (iii)	3	14.3E	HE3
14 (a)	3	2.9E, 2.10E	PE3
14 (b) (i)	3	17.3	HE7
14 (b) (ii)	2	17.3	HE7
14 (c) (i)	1	2.3, 2.5	PE2, HE7
14 (c) (ii)	1	2.3, 2.5	PE2, HE7
14 (c) (iii)	4	7.3	PE2, PE6, HE7
14 (c) (iv)	1	1.4, 2.3, 2.5	PE2, PE6, HE7